1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICO AND GNSS
J1 pad n = ISO A pin n. A1 REVERSE A2 HANDBRAKE A3 AMP REMOTE A4 BATT+ A5 remote (R57) A6 ILLUM A7 ACC A8 GND
Always-on 5.2V: MCU VSYS + GNSS LDO. TPS560430X runs PFM at light load (~55uA Iq).
2. VEHICLE INPUTS (opto-isolated). ACC, ILLUM, REVERSE: +12V = on. PARK: switch to ground = on.
ACC (ISO A7)
ILLUM (ISO A6)
REVERSE lamp (ISO A1)
PARK (ISO A2, handbrake switch; may stay unwired)
LED ~2.7mA at 14.4V; EL817 C-grade CTR >= 200%. 4.7k pull-downs satisfy RP2350-E9.
3. AMPLIFIER REMOTE (+12V from ACC, MCU GP9)
Goes to ISO A3 and C1-6 (A5 via R57). ~1A capable; remote inputs draw <100mA.
Only works while ACC is on, even if the MCU hangs.
4. MCU HELPERS (MCU core is section 13)
GPIO map: see section 13 and DESIGN_rev03.md.
5. GNSS: u-blox NEO-M9N, active antenna on SMA
Active antenna (3.3V): VCC_RF -> 10R -> 27nH onto RF_IN. RF_IN is DC-blocked inside the module (verify in the integration manual).
6. VIM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the MCU)
Display buttons (to GND): GP16 screen, GP17 vol+, GP22 vol-, GP27 mute. J12.
Backlight and reverse moved off the VIM3 header; the VHAL now sets/reads them through the MCU.
7. AUDIO: VIM3 TDM-B I2S -> 2x PCM5102A (front + rear) -> line out on ISO C1
2.1 Vrms ground-centred line out, no coupling caps. 24-bit in 32-bit I2S slots, 48/96/192 kHz.
I2S series resistors at the ribbon entry.
8. VIM3 40-PIN HEADER + ISO 10487 C1 (line out)
J8 pad n = ISO C1 pin n: 1 RL, 2 RR, 3 FL, 4 FR, 5 line GND, 6 remote +12V.
VIM3 pin v -> IDC pad 2v-1 (v<=20) or 2(v-20) (v>20). Pin 1 sits opposite pin 21 (confirmed by Glenn).
Pin 35 (TDM-B lane 3) left free for a later sub/3rd pair. MCLK (30) not used.
Car radio peripheral board rev 0.3 - MG F (1996), VIM3 + RP2350B, 4ch line out, NEO-M9N. Generated; see DESIGN_rev03.md.
9. DISPLAY: Waveshare 7in 70H-1024600 adapter built in (HDMI in from VIM3 -> 40-pin FPC to the panel)
Panel 5V (~0.45A) comes from the board's switched 5V rail (+5V_SYS, section 10), not from VIM3 USB. Touch USB goes to the on-board hub (section 11).
Backlight EN/PWM now go straight to the panel: 0V = brightest, 3.3V = darkest (invert PWM in firmware). J6 removed.
FPC orientation as on the Waveshare adapter: seen from the top, ribbon leaving the board edge, pin 1 is on the RIGHT, pin 40 on the left.
10. 5V/5A SYSTEM SUPPLY FOR VIM3 + SCREEN + HUB (TPS55288 buck-boost, rides through cranking and overvoltage)
Input 3V-36V (runs through cranking); output 5.0V up to 5A. MCU GP28 = EN, MCU I2C0 sets OE/voltage/current limit.
VIN is permanent +12V so the VIM3 can finish its Android shutdown after ACC drops.
11. USB HUB: VIM3 header USB (pins 3/4) -> MCU, screen touch, spare port
One link to the VIM3: no USB cables. Port 1 MCU (HID + GNSS CDC), 2 touch, 3 RTL-SDR, 4 second hub (external ports). Route D+/D- as 90 ohm pairs.
VIM3 header pins 3/4 are a working USB host port (confirmed by Glenn, 2026-09-29).
12. EXTERNAL USB PORTS (phone etc.): second hub + dedicated 5V/3A supply, 1.5A current-limited per port
Ports are powered only while the VIM3 supply is on. All USB shares the VIM3 header's USB 2.0 link (480 Mbit/s); RTL-SDR needs ~40 Mbit/s.
13. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design
GPIO0-29 as rev 0.2 except: GPIO40 battery ADC (was GP26), GPIO23/24 USB port faults, GPIO25 LED, GPIO26 USB supply PG,
GPIO29 GNSS safeboot, GPIO30/31 on J10, GPIO32-39 on J21, GPIO41 5V rail ADC, GPIO42 board temperature, GPIO43 hub 1 reset.
RP2350-E9 check (every MCU pin): opto inputs GP2/3/6/7 4.7k down; SBC sense and service jumper 4.7k down; buttons, USB faults/PG, I2C, RUN pulled up;
GNSS PPS/RX 4.7k down (R94/R95); outputs have defined off-state resistors; ADC pins run with the digital input disabled; unused and expansion pins: see DESIGN_rev03.md.
Sheet: /
File: carradio_peripheral_rev03.kicad_sch
Title: Car radio peripheral board (RP2350B + VIM3)
Size: A0 Date: 2026-09-29 Rev: 0.3
KiCad E.D.A. 10.0.6 Id: 1/1